ARDUINO: I2C PROTOCOL & CONNECTING MULTIPLE DEVICES

Learning how Arduino talks to many devices

Let's Discover!

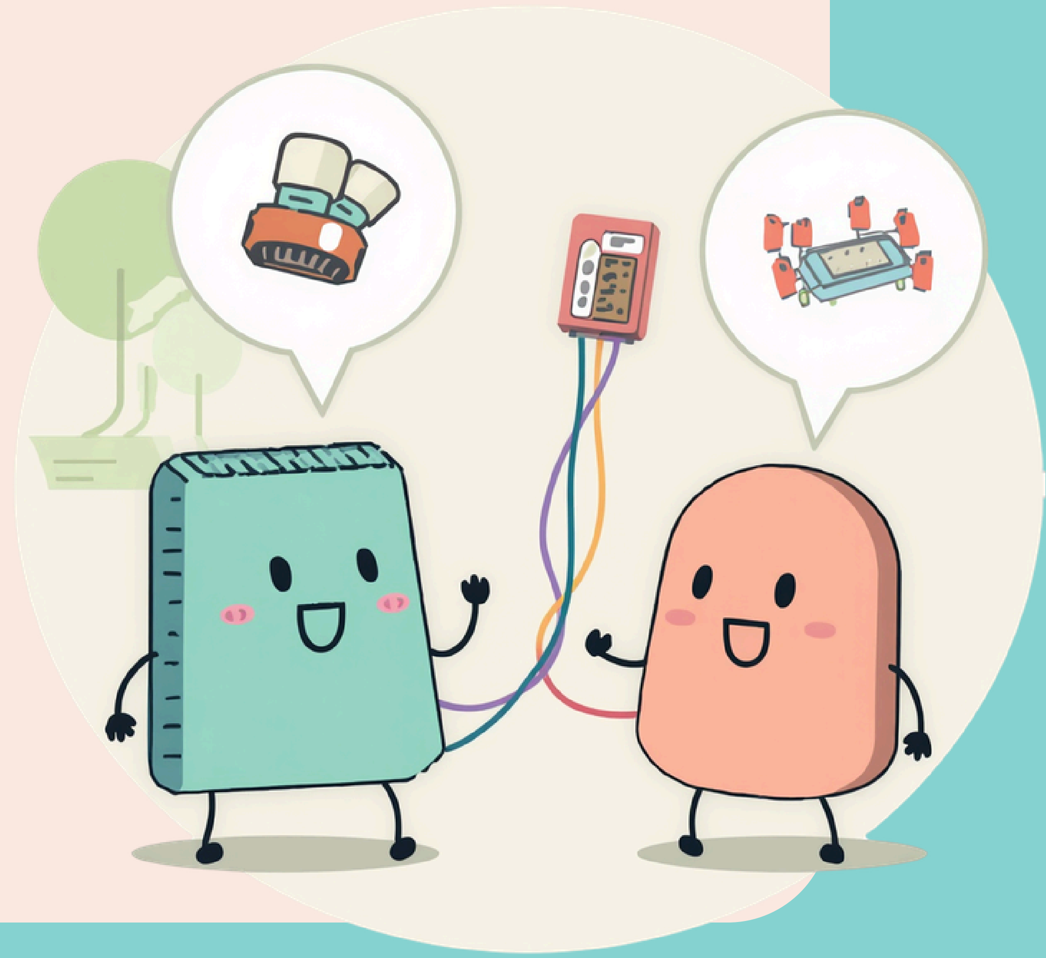
What is Arduino?

Arduino is a tiny computer that helps you control electronics! Think of it like a brain for your projects. You can make lights blink, sensors detect things, and motors move. The best part? Arduino can talk to many devices at once, making amazing projects possible!

How Do Devices Talk to Arduino?

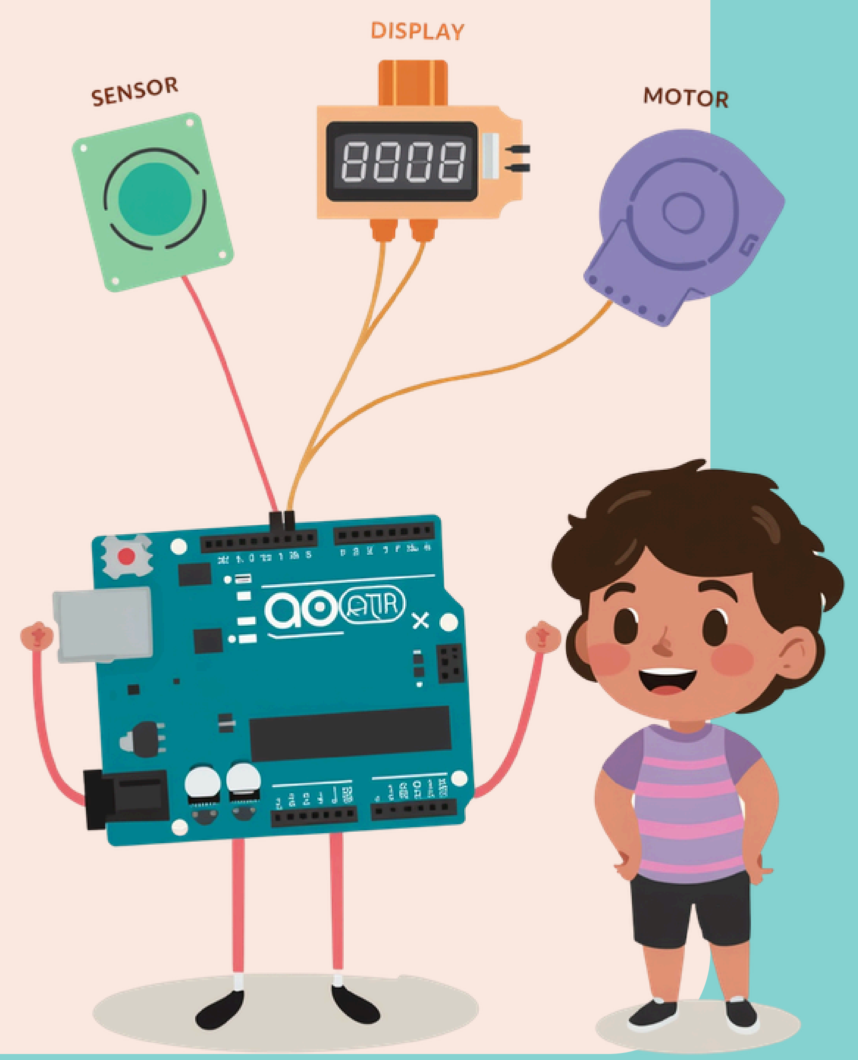
Devices use wires to send messages back and forth!

Key Terms: Communication & Protocol



Meet I2C: The Friendly Device Talker

I2C is a special way for Arduino to talk to many devices using just two wires!
Fun Fact: I2C stands for "Inter-Integrated Circuit"



Why Use I2C?

Saves Wires

Easy to Connect

Works with Many Devices

How Does I2C Work?

SDA (Data) + SCL (Clock)

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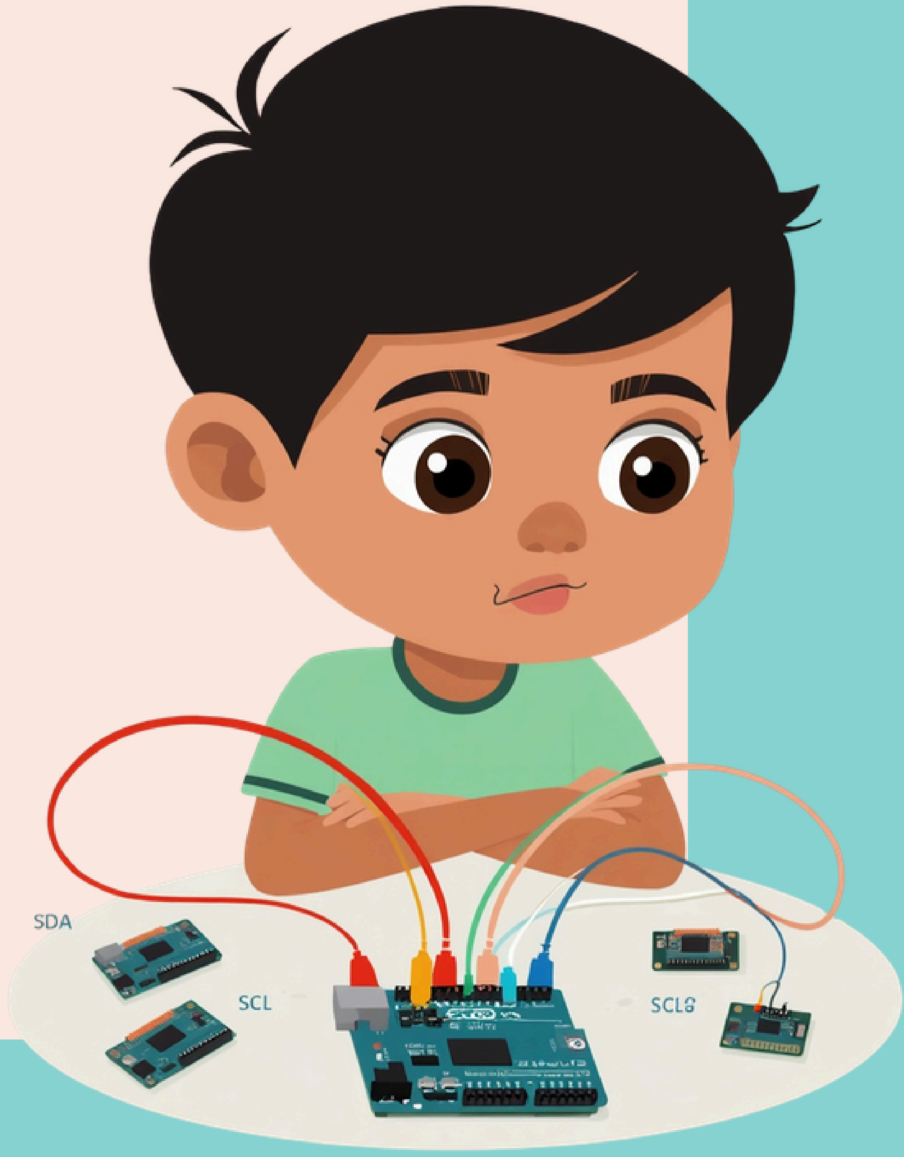


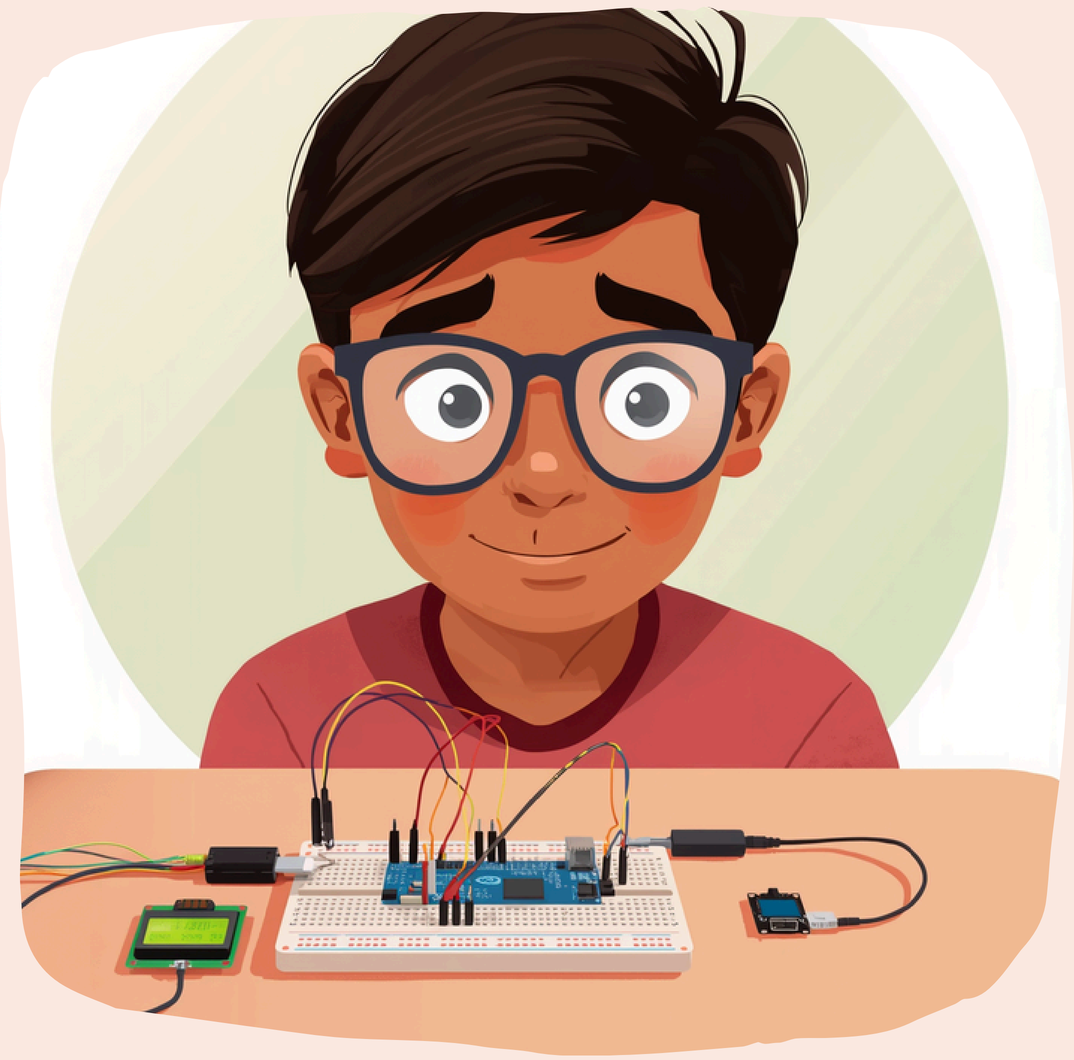
I2C Addresses: Who is Talking?

Sensor - 0x40

Display - 0x27

Motor Controller - 0x60

Connecting Multiple Devices with I2C



I2C Connection Demo

See how many devices share the same two wires! With I2C, your Arduino can connect to sensors, displays, and other devices all at once. Each device gets its own address, so Arduino knows exactly who to talk to. No more messy wires everywhere!

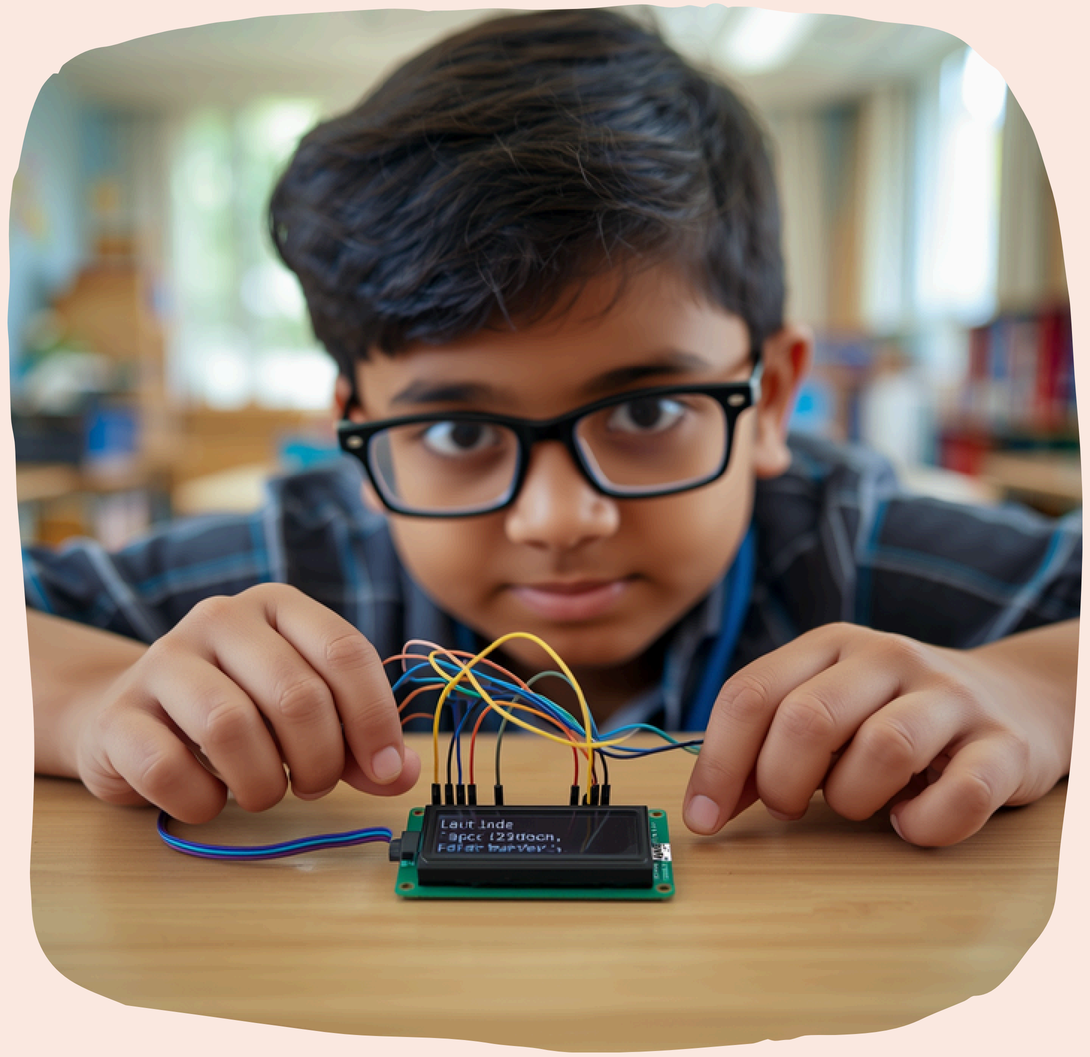
Example 1: Connecting Sensors

Temperature Sensor - Measures heat

Light Sensor - Detects brightness

Accelerometer - Senses movement

Example 2: Connecting Displays

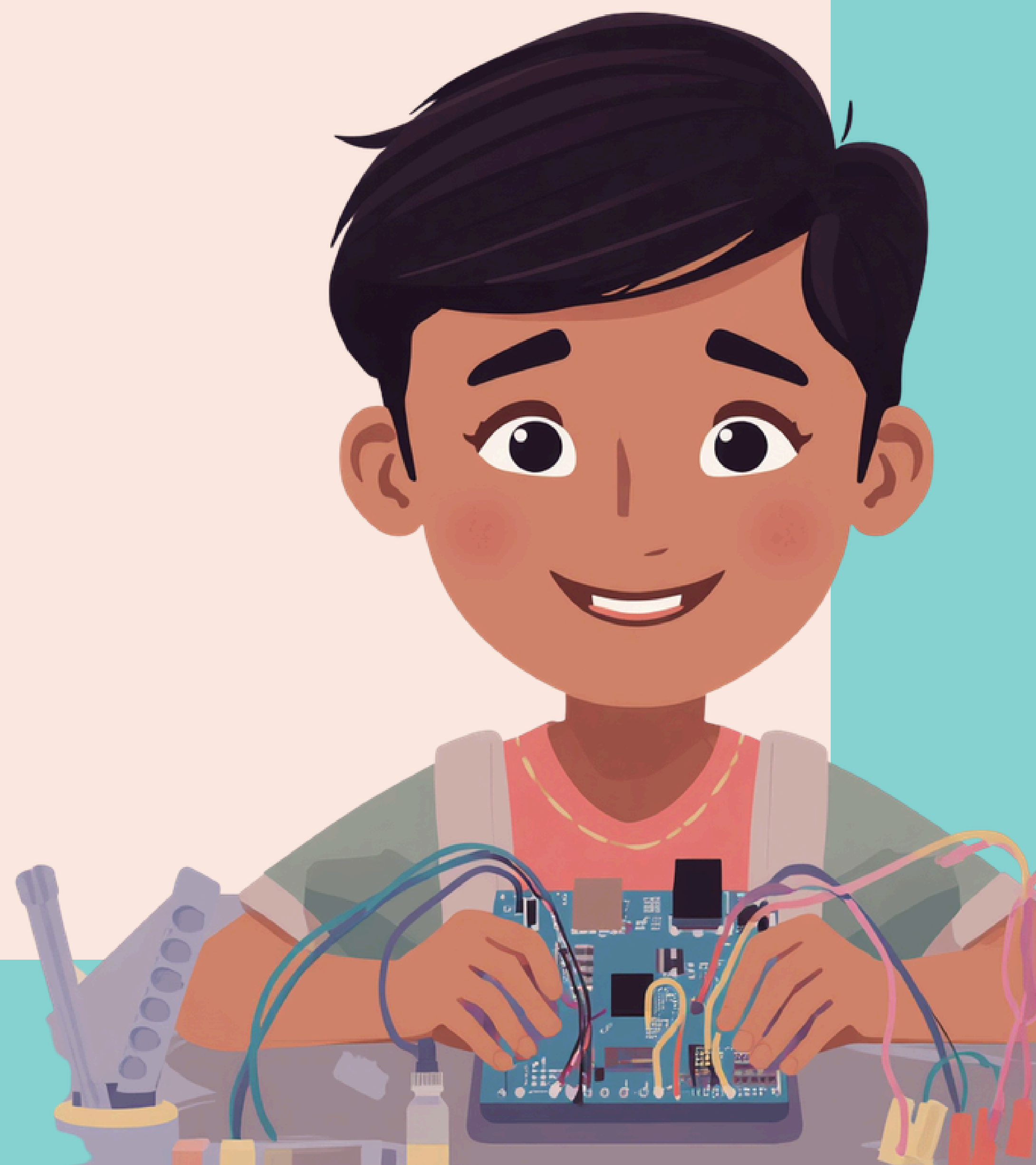


I2C Display Demo

Displays show information from Arduino easily with I2C! With just two wires (SDA and SCL), you can connect LCD or OLED screens to show sensor readings, messages, or fun graphics. It's like giving your Arduino a voice to share what it knows!

Hands-On Activity: Build Your I2C Circuit!

1. Gather Arduino & I2C devices (sensors or display)
1. Connect SDA & SCL wires to all devices
1. Upload code & watch devices work together!



Tips for Working with I2C

Connect SDA and SCL wires correctly

Check device addresses carefully

Keep wires tidy for best results

Fun Facts About I2C

"I2C was invented in 1982 by Philips to help chips talk easily!"

Over 100 types of devices use I2C today!

What We Learned About I2C

I2C uses just two wires to connect many devices
Each device has a unique address to communicate
Great for sensors, displays, and more projects!

Your Turn!

Questions & Ideas

Thank You & Keep Exploring!